KENNETH SAUERS

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EXPERIENCE

Software Engineer, System Innovation Group **Dec 2020 – Current**

- Developed a high-accuracy machine learning classifier (98%) to detect base stations, strengthening network security.
- Spearheaded the consolidation of multiple applications into a unified framework, streamlining deployment and accelerating prototyping.
- Authored and released comprehensive technical documentation including Interface Control Documents (ICDs), design specifications, and tooling guides for internal teams and external clients to ensure seamless integration and support.
- Collaborated with stakeholders to deploy rapidly prototyped solutions tailored to real-world operational and engineering problems.

Software Engineer, Department of Defense - NSIN **May 2020 - Sep. 2020**

- Collaborated one-on-one with active-duty airmen to develop a simulation tool for KC-135 aircraft maintenance, enhancing training realism.
- Worked with domain experts in engineering processes to translate requirements into technical prototypes.
- Designed and prototyped a VR application to train aircraft maintainers, improving hands-on learning experiences.

Research Assistant, Center for computer vision research lab UCF **Sep. 2017 - Mar. 2020**

- Developed a synthetic data generation tool for machine learning, increasing testing accuracy by 13%.
- Contributed to research on Generative Adversarial Networks (GANs) and their applications in Computer Vision, enhancing data augmentation and model performance.
- Researched and applied deep learning architectures, including CNNs and LSTMs, to improve visual recognition tasks.

EDUCATION

Master of Science in Computer Science, Georgia Institute of Technology **May 2025**

- Specialization in Computational Perception and Robotics
- Coursework in Robotics Algorithm, Computer Vision, NLP, Deep Learning, and Reinforcement Learning

Bachelor of Science in Computer Engineering, University of Central Florida **May 2021**

- Minor in Intelligent Robotic Systems
- Robotics Team, Computer Vision Engineer
- Competitive Programming Team, Programmer

PROJECTS

Personal Project | BDLC Actuator **2024**

Developed a robotic actuator with Field-Oriented Control (FOC) for precise position, velocity, and torque regulation via a CAN interface.

Swamphacks | Skin Disease Diagnosing Mobile App **2020**

A computer vision app that detects skin disease via a convolutional neural network

- **Award**, Most Technically Impressive
- **Technology**, Convolutional Neural Network, OpenCV, Pillow, Native Android, Flask, and GCP

SKILLS

- **Machine Learning & AI**: CNNs, RNNs, LSTMs, Reinforcement Learning, GANs, NLP
- **Certifications** Coursera: Deep Learning Specialization, CIW: JavaScript Specialist
- **Languages** Python, C#, TypeScript, JavaScript, Java, C++, SQL
- **Frameworks** .Net, ASP.NET, Entity Framework, React, ROS, TensorFlow, PyTorch, OpenCV
- **Technology** Docker, DevOps, CI/CD, SQL, GnuRadio